# Subashanan Nair

Data Scientist

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github.com/SubaashNair | subashanan-nair.vercel.app

## PROFESSIONAL SUMMARY

Data Scientist with 7+ years of experience across machine learning, data engineering and technical education. Builds production data pipelines, dashboards and LLM powered internal tools for Penang Institute and Forward College. Hands-on with Python, scikit-learn, TensorFlow, Streamlit, Docker and GitHub Actions. DataCamp certified AI Engineer for Data Scientists and Kaggle Top 100 competitor.

## SKILLS

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|---|---|
| **Languages** | Python, SQL, R, JavaScript, TypeScript |
| **ML & AI** | scikit-learn, TensorFlow, Keras, XGBoost, LightGBM, CatBoost, Deep Learning, Reinforcement Learning, LLM applications (Ollama, prompt design, RAG, text to SQL) |
| **Data & MLOps** | Pandas, NumPy, Streamlit, Plotly, Matplotlib, Seaborn, MLflow, Docker, GitHub Actions, Tableau, Google Drive API |
| **Web & Databases** | Next.js, FastAPI, Node.js, REST API design, PostgreSQL, Supabase, Quarto |
| **Tools** | Git, GitHub, Jupyter Notebook, Streamlit Cloud |

## EXPERIENCE

**Forward College** | *Instructor & Data Scientist* Feb 2019 to Present

- Instruct 3 cohorts per year in Python, machine learning and data analytics, averaging a 4.7/5 learner satisfaction score.
- Built internal programme tools: an LLM assisted project review dashboard, a GitHub based submission and assessment platform, and an exam monitoring app.
- Engineered and deployed ML pipelines and predictive models with Python, scikit-learn and MLflow for client organisations, reducing manual analysis time by 60%.
- Delivered data solutions and dashboards for 3+ organisations, including Penang Institute, enabling data driven policy decisions.

**Penang Institute** | *Data Scientist (Consultant)* Jun 2024 to Present (2 yrs 4 mos)

- Diagnosed a stale public statistics site and replaced a 34 script R workflow with an automated Python pipeline (GitHub Actions, Docker) that publishes OpenDOSM and data.gov.my data to Google Drive for Tableau.
- Built a weekly ingestion pipeline for key Penang indicators (population, GDP, employment, CPI, household income) backed by a 31 test validation suite.
- Created a pull request gated Streamlit uploader so staff can add new publications without touching code, and an interactive key statistics dashboard (Streamlit, Plotly) deployed on Streamlit Cloud.
- Maintained the Quarto statistics website and delivered a client requested website revamp.

**Kuen Lai International Secondary School** | *STEAM Consultant* May 2013 to Jan 2019

- Taught Python programming, Arduino and Raspberry Pi electronics, and 3D printing through hands-on, project based STEAM lessons.
- Designed and delivered STEAM programmes aligned with national and international curriculum frameworks.
- Ran workshops for educators on STEAM pedagogy and digital literacy.

## PROJECTS

**Forward College Project AI Review Tool** | *Internal Tool*

*Python, Streamlit, GitHub API, Ollama Cloud (gpt-oss:120b), GitHub Actions*

- Monitors 9 student capstone repositories and scores each against a 10 section rubric. The LLM reviewer must cite exact source quotes as evidence; fabricated quotes are rejected and the final percentage is computed locally, not taken from the model. Runs hourly on GitHub Actions and never executes student code.

**FC Submission Platform** | *Internal Tool*

*Next.js, TypeScript, Supabase, GitHub OAuth, Playwright*

- Students register a public GitHub repo; the platform snapshots the commit at the deadline and runs an automated deliverable checklist. Instructors can override results and grant exceptions with a full audit trail. LLM rubric grading is specified for the next phase.

**Exam Guard, Exam Monitoring App** | *Internal Tool*
*Rust, Tauri, TCP/UDP networking, GitHub Actions*

- LAN proctoring tool needing no accounts or internet: a live grid of student screens with problem tiles surfaced first, UDP discovery and class codes. Ships cross platform releases for Windows, macOS and Linux.

**Penang Economic Data Assistant** | *Prototype*
*FastAPI, PostgreSQL, Next.js, Docker, Ollama Cloud*

- Natural language assistant over OpenDOSM CPI and unemployment data. Questions are translated to governed SQL with an audit trail, alongside a Next.js data explorer.

**Penang Pulse** | *Spatial Analysis*
*Python, NumPy, Pydantic, Penang GeoHub, OpenStreetMap, DOSM data, Next.js, React Three Fiber*

- Reproducible study of how evenly clinics, hospitals, schools, police stations, fire stations and markets are distributed across Penang districts, expressed as service density per 10,000 residents. Limitations are published alongside every result and artefacts carry SHA-256 checksums.

**Forecasting Lab** | *Predictive Modelling*
*Python, scikit-learn, NumPy, SciPy, Pydantic, UCI dataset 608*

- Leakage safe 15 minute traffic forecasting benchmark across a 36 sensor network. Histogram gradient boosting cut MAE by 31% versus a persistence baseline and a spatial MLP cut it by 22%.

**LLM Eval Lab** | *In Progress*
*Python, LLM APIs, open weight models, bootstrap confidence intervals*

- Designing a benchmark of a classical baseline, a fine tuned 1B model, an open 8B model and a frontier model on Manglish sentiment, Bahasa Malaysia entity extraction and summarisation, reporting accuracy with 95% intervals, cost and latency from hand labelled data.

**Documentation MCP Server**
*JavaScript, Node.js, Docker, REST API*

- Aggregates live documentation from open source libraries (React, Vue, Angular and more) into a searchable REST API and web UI, with daily cron updates and Docker Compose deployment.

## EDUCATION

**Asia e-University** | *Master of Education in Science* May 2014 to Oct 2016
**Universiti Tun Abdul Razak** | *Bachelor of Education (Hons)* May 2009 to Sep 2012

## CERTIFICATIONS

**DataCamp** | *AI Engineer for Data Scientists Associate* Apr 2026
*Certificate ID: AEDS0019212905886*
**Coursera (DeepLearning.AI & Stanford)** | *Supervised ML: Regression and Classification* 2024
**Google for Education** | *Google Certified Educator Level 1* Dec 2021
**Kaggle Competition** | *Top 100 placement in a global machine learning competition*